Capstone Synthesis Paper - Healthcare Appeal Letter Generator with Public Decision Analytics

1. Introduction

My professional background is in healthcare including experience working in provider offices and various health insurance companies. Through these experiences, I have developed an understanding of healthcare operations, insurance processes, and the challenges faced by patients, providers, and health plans. Healthcare can be difficult to navigate and fully understand especially when individuals are simultaneously dealing with medical concerns and coverage decisions. Denials and appeals can create additional confusion, frustration, and administrative work for everyone involved. From my experience on the insurance side, there is also a need for better data surrounding appeals to identify patterns, evaluate processes, and determine where improvements may be needed.

One of my primary reasons for pursuing a Master of Science in Computer Information Systems and Business Analytics (MS-CISBA) was to broaden my knowledge and develop stronger technical and hands on skills in information systems and business analytics. I wanted to better understand how these areas work individually and together to address business goals particularly in the healthcare industry. Data and technology are increasingly important in healthcare particularly for improving operational efficiency, supporting business decisions, identifying patterns, forecasting, and developing innovative solutions.

For my final capstone, I chose the Healthcare Appeal Letter Generator with Public Decision Analytics as it brings together my professional healthcare experience and the knowledge and skills I have gained throughout the MS-CISBA program. It represents a culmination of my experience, education, and interest in addressing a real healthcare problem. My goal was to develop something practical that could demonstrate what I have learned while also providing an opportunity for expansion and innovation.

This project focuses on supporting the healthcare appeals process through public decision data and an editable appeal letter draft. A healthcare appeals coordinator can use the prototype to review historical decision statistics and prepare a draft using case information. The larger idea is that data and technology could eventually help make appeals more understandable and manageable while supporting more efficient and informed decisions for healthcare organizations, and create opportunities for innovation within this area.

I intentionally kept the prototype manageable due to the limited timeframe and my current technical experience. Although I understand the broader concepts behind software systems, databases, cybersecurity, and analytics, I am still developing my ability to implement more complex technical solutions independently. Rather than attempt a larger application, I focused on creating a working prototype that connects several curricular areas through one complete process. Although limited in scope, it provides a foundation for a more comprehensive healthcare appeals support system which I explore in my one year vision below.

2. The Four Curricular Areas

The MS-CISBA program brings together four main curricular areas: Data Analytics, Data Mining and Management, Systems Development, and Networking and Cybersecurity. I understand these areas as different but connected parts of an information system. Each contributes something necessary to turn data and a business need into a useful and functional application.

Data Analytics

Data Analytics involves taking data and turning it into meaningful information that can support decisions. This includes preparing data, identifying patterns, calculating results, and presenting information in a way that users can understand. The purpose is not simply to produce statistics or visualizations, but to determine what the information means and how it can be useful.

Through CIDM 5310 and CIDM 6308, I gained experience with data preparation, business intelligence, visualization, and analytical tools such as Python, Streamlit, Tableau, Power BI, and RapidMiner. My CIDM 5310 Stock Price App demonstrates experience developing an interactive application while my CIDM 6308 regression artifact provides evidence of predictive analytics coursework. As discussed in my Data Analytics appraisal, I understand how these tools support business decisions although I am continuing to develop my ability to build more advanced analytical solutions independently.

In my capstone, Data Analytics is demonstrated through historical California Department of Managed Health Care (DMHC) Independent Medical Review decision statistics. The application filters records by treatment category and review type, calculates the number of decisions that upheld or overturned a health plan's determination, and displays the results for the user.

Data Mining and Management

I understand Data Mining and Management as the processes used to collect, organize, store, prepare, and retrieve data so that it can be used reliably. Data must be accessible and properly structured before meaningful analysis can take place. This area also involves identifying data quality problems and determining how information should be maintained. Data mining builds upon these processes by examining datasets to identify useful patterns and relationships that may not be immediately apparent.

CIDM 6350 introduced me to relational databases, SQL, data modeling, and normalization. My database schema artifact demonstrates my experience with database design. CIDM 6351 expanded my understanding of ETL processes, data warehouses, and data pipelines. My PostgreSQL coursework artifact provides evidence of SQL query work using pgAdmin. I also studied Snowflake concepts although my practical database coursework involved PostgreSQL rather than a full Snowflake implementation.

In the capstone, Data Management is demonstrated through retrieving the official DMHC CSV, checking its records, and importing selected fields into SQLite. The application then uses these records for the historical analysis portion.

Systems Development

Systems Development involves designing and building applications that addresses a need and allows users to interact with data and functionality. I understand this area as the process of turning a problem and a set of requirements into an actual working system.

Of the four curricular areas, Systems Development remains my weakest technical area. CIDM 6325 introduced me to web development, application structure, GitHub, and development workflows. My earlier HealthAppeals coursework project provided exposure to Django models, views, forms, and templates. That experience gave me a foundation for understanding how the different parts of a web application connect.

As explained in my Systems Development appraisal, I am stronger at understanding what a system needs to accomplish than independently implementing every technical component. Developing this capstone allowed me to apply that understanding while gaining additional experience with Python, Django, troubleshooting, and testing.

Systems Development is demonstrated through the working Django application. The user selects a treatment category and review type, enters fictional case details, views historical statistics, and generates an editable appeal letter draft.

Networking and Cybersecurity

I understand Networking and Cybersecurity as the area responsible for protecting information, connections, and applications from risk. This includes network configuration, identifying vulnerabilities, evaluating threats, and determining appropriate safeguards.

CIDM 6340 and CIDM 6341 provided experience with security assessments, auditing, risk identification, and tools such as NMAP, Wireshark, and ShieldsUP. My CIDM 6340 network audit plan and CIDM 6341 security report demonstrates this coursework. My Networking and Cybersecurity appraisal also explains that I understand security risks from an organizational and healthcare perspective while advanced technical implementation remains an area for development.

Networking and Cybersecurity are particularly important in healthcare as information must be protected and the consequences of improper access or disclosure can be significant.

The current capstone prototype applies basic security principles through input validation, responsible data handling, and the use of fictional case information. It does not connect to an external AI service or require real patient records. However, it is not a production application and does not automatically prevent a user from entering sensitive information. A larger system

involving actual healthcare information would require substantially stronger security, access controls, and organizational oversight.

3. How the Areas Work Together

The purpose of my capstone is to demonstrate how these curricular areas can work together. The prototype integrates Systems Development, Data Management, and Data Analytics through one complete workflow while Networking and Cybersecurity guides the design and limitations.

The process begins with the official California DMHC Independent Medical Review dataset. This public dataset contains historical decisions about services reviewed through California's independent medical review process. It provides actual decision outcomes that can be analyzed for historical context. The original CSV is downloaded from the official source using a script and imported into a SQLite database.

Once the records are stored in SQLite, the Django application allows the user to select a treatment category and review type. The application retrieves matching decisions, calculates the historical counts, and displays the results along with three example source reference IDs. For example, selecting Pharmacy and Medical Necessity returns 10,646 historical decisions**,** including 4,071 upheld decisions and 6,575 overturned decisions. The matching example source reference IDs are MN01-594, MN01-606, and MN01-607**.**

The user can then enter fictional denial information, a medical necessity explanation, and supporting document information. Django uses these details and the historical statistics to create an editable appeal letter draft from a predefined Python template.

This workflow demonstrates the relationship between the different curriculum pillars. Without Data Management, the application would not have organized public records available for analysis. Without Data Analytics, it could not calculate and present the filtered historical results. Without Systems Development, there would not be an interactive application connecting the information to the appeal drafting process. Networking and Cybersecurity also have an important role even though the prototype does not demonstrate a comprehensive security infrastructure. The use of fictional case information, basic form validation, and avoidance of unnecessary external services reflect security considerations within the limited scope.

4. Design Decisions, Testing, and Feasibility

A major decision in this project was to focus on a smaller working prototype instead of attempting every feature I could envision. My interest in healthcare appeal letter generation was influenced by my healthcare background and earlier HealthAppeals coursework. However, the capstone is a separate Django application that elaborates on prior coursework and connects real public decision data to a letter drafting process.

As for data, I decided to utilize California's public DMHC dataset per the professor’s suggestion and because it provides real historical decisions that can be retrieved and examined. This allowed me to demonstrate data integration and analysis without relying on synthetic decision records.

I also chose SQLite as it provides a manageable database for the prototype. My coursework introduced more complex database technologies. However, SQLite allowed me to focus on understanding and connecting the essential components before expanding the architecture.

The letter generator uses a predefined template rather than an artificial intelligence service. This was another intentional scope decision. A template provides a transparent and editable starting point while avoiding the added complexity of integrating an external language model, evaluating its output, and managing potential privacy concerns.

The application is not intended to provide legal or medical advice, automatically submit an appeal, or predict the outcome of a particular case. Historical appeals decisions also do not represent every type of health insurance denial or appeal, thus the results must be interpreted within these limits.

To verify the prototype, I developed four automated Django tests covering input validation and the accuracy of the historical statistics and source reference IDs. I also recreated the application from a fresh GitHub clone including installing its dependencies, creating a new database, downloading the complete official CSV, and importing all 42,710 DMHC records**.** All four tests passed and the historical Pharmacy and Medical Necessity tests results matched the original application. This confirmed that the project could be reproduced using the documented instructions.

The project also reflects both my existing strengths and areas for growth. My healthcare experience helped me identify the problem and understand why an appeal support tool could be useful. My analytics and data management coursework provided a foundation for working with public data, while my systems coursework helped me understand the application components. The capstone gave me practical experience connecting these areas and additional opportunities to develop technical skills.

For these reasons, I consider the prototype a feasible synthesis. It demonstrates that public healthcare decision data, historical analysis, and an interactive appeal drafting process can work together in one application.

5. One Year Vision

My larger vision is to expand the prototype into a more comprehensive healthcare appeals support system. I see value for provider offices, healthcare appeals coordinators, insurance companies, and individuals who need help understanding and preparing appeals.

These groups share a connection to the appeals process, but they have different responsibilities and information needs. A provider office may need help preparing documentation and tracking appeals. An insurance company's appeals or finance department may want to identify patterns and improve internal processes. An individual may need clearer guidance about required information, deadlines, and how to prepare a letter during an already stressful time. A full system would need to recognize these differences rather than provide the same interface and workflow for every user.

Expanding Data Management

I would research public appeals or independent review datasets from additional states and determine whether they provide information that could be used the same way. Since states may collect and categorize information differently, the data would need to be evaluated before combining it or presenting comparisons.

The current Python import process provides a foundation for this work. Over one year, it could be expanded into a more automated data pipeline that retrieves updated datasets, validates records, standardizes useful fields, and records the source and date of each update. A more structured database such as PostgreSQL could support additional data sources and more complex queries.

Developing Interactive Analytics

The current prototype displays historical counts and source reference IDs. I would like to expand this into a more visually appealing application with interactive dashboards, charts, filters, and historical trends.

Different users could access information relevant to their needs. For example, a healthcare appeals coordinator could review historical outcomes by treatment category and review type. An organizational user might examine patterns over time or compare categories to identify areas that require further review.

My experience with Tableau, Power BI, and RapidMiner provides a foundation for understanding these possibilities. The prototype has already demonstrated that the public records can be filtered and used to generate meaningful descriptive statistics. The next step would be presenting those results more effectively and developing stronger methods for evaluating their quality and usefulness. Any forecasting or more advanced analysis would require additional development and validation.

Improving the Application and Letter Generator

Additionally, the website could be more visually appealing, easier to navigate, expanded functionality, and tailored to different types of users. For healthcare coordinators and provider offices, the application could include organized appeal workflows, document checklists, and tools to help manage required information.

The letter generator could also become more specific to individual health insurance companies. For example, it could use a maintained directory of verified insurer addresses, appeal instructions, and document requirements to populate appropriate information automatically. This would reduce repetitive work and help users prepare drafts. The system would still need to allow users to review the information for final approval.

Another possible expansion is connecting to internal data for authorized healthcare users. This would require permission, appropriate technical integration, and privacy protections. It would not be a simple extension of the current prototype.

The existing Django application demonstrates the basic interaction between user input, database queries, historical statistics, and an editable letter. Those working components provide a starting point for building more advanced interfaces and workflows.

Strengthening Networking and Cybersecurity

Networking and Cybersecurity would become a more substantial part of the full system, especially if it were made available to multiple users or connected to healthcare organizations.

A public informational tool using public decision data has different security needs from an organizational system containing actual patient or insurance information. Before allowing the use of real patient information, I would need to address authentication, user permissions, secure storage, audit logging, data protection, and applicable privacy requirements. Connections to healthcare organizations would also require authorized access and appropriate agreements.

I would begin with secure architecture and access controls before expanding to sensitive organizational data. This reflects what I learned in CIDM 6340, CIDM 6341, and CIDM 6375 about evaluating risks and considering the organizational consequences of security decisions.

Bringing the Full System Together

I would approach the one year development process in stages. The initial work would focus on improving the existing application, strengthening the data pipeline, and creating more useful dashboards. The next stage would introduce customizable letter templates and workflows tailored to healthcare coordinators and provider offices. Additional public state datasets could be incorporated where the data is available and comparable.

More advanced organizational features including connections to insurance company systems or handling real patient information would depend on the security, authorization, and technical requirements involved. These could become a later stage rather than features added before the necessary foundation exists. This larger vision would require me to continue developing the technical skills I identified as weaknesses in my curricular appraisals particularly advanced Python programming, database development, software architecture, and cybersecurity implementation.

Ultimately, I chose this project as it brings together everything I hoped to gain from the MS-CISBA program and connects it to the healthcare industry where I have spent much of my professional career. The current prototype is intentionally limited, but it demonstrates a complete process that works. My one year vision builds upon that process to explore how data, technology, and business analytics could make healthcare appeals more understandable and manageable while supporting more efficient and informed decisions for the different parties involved.